

Intro to Python

Material adapted from previous workshops by
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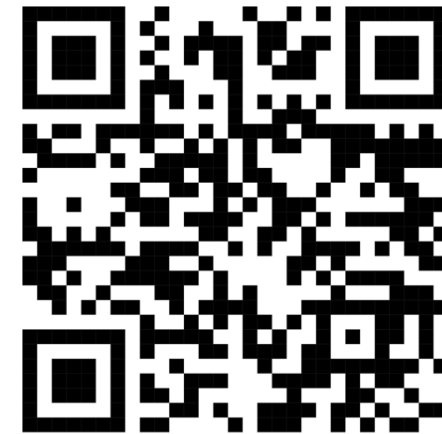
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Workshop Series

Workshop	Date	Location	Registration
How to Think in Code	May 26 , 11 AM - 1 PM	Sherbrooke 550 Rm 137	<u>Open</u>
Intro to Git & GitHub	May 27 , 11 AM - 1 PM	Sherbrooke 550 Rm 137	<u>Open</u>
Intro to Unix	May 28 , 1 - 3 PM	Sherbrooke 550 Rm 137	<u>Open</u>
Intro to Python (Part 1)	June 3 , 11 AM - 1 PM	Sherbrooke 550 Rm 137	<u>Open</u>
Intro to R (Part 1)	June 5 , 11 AM - 1 PM	Sherbrooke 550 Rm 137	<u>Open</u>
More workshops to be announced soon!			

<https://www.mcgill.ca/micm/training/workshops-series>



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Learning Outcomes

Summary

In this 2-hour workshop, participants will be introduced to the basics of programming in Python. Students will journey from the beginnings of creating variables and performing simple mathematical operations to writing code that can perform fundamental tasks and wrapping this code into functions. Participants will learn how to write the important building blocks that make up complex programs.

Learning Objectives

1. Store data in variables and collections.
2. Perform basic operations on these data.
3. Use control flow and loops to write powerful code.
4. Define functions to create repeatable units of behaviour.



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Outline

1. Module 1 – Python Basics (30 minutes)

- a. Hello, World!
- b. Variables
- c. Numbers and Comparisons
- d. Intro to strings
- e. Exercise**

2. Module 2 – Collections (30 minutes)

- a. Lists and List Methods
- b. Tuples
- c. Dictionaries
- d. Exercise**



Outline

- 3. Module 3 - Intro to Control Flow and Loops (30 minutes)**
 - a. Control Flow: the if statement
 - b. Loops
 - c. Exercise**
- 4. Module 4 – Introduction to Functions (25 minutes)**
 - a. Function Overview
 - b. Writing Custom Functions
 - c. Documenting Functions
 - d. To script, or not to script?
 - e. Exercise**
- 5. Module 5 – Where to go from here (5 minutes)**



Interactive Workshop!

- That's pretty much all that will be in the slides... For the rest, we'll go to a Jupyter Notebook:



To the repository!



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To summarize

- ✓ Data can be stored in **variables** of several types, including **strings**, **integers**, **floating point numbers** and **Booleans**.
- ✓ **Collection types**, such as **tuples**, **lists** and **dictionaries** can be used to store **multiple** data points.
- ✓ **Control flow** and **loops** help decide which lines to run and allow lines to be repeated.
- ✓ **Functions** help package up behaviour into units that you can easily reuse.

Now you are ready to:

- Store data in variables and collections.
- Perform basic operations on these data.
- Use control flow and loops to write powerful code.
- Write functions to repeat complicated tasks.



Acknowledgements

- Thank you to QLS-MiCM for giving me this opportunity and for helping me along the way.
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- Thank you to the Python community!



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Feedback



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